

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz 2
Semester: Spring 2025

Duration: 20 min
Full Marks: 10

CSE 370: Database Systems

Name:

ID:

Section:12

Design a database to keep track of information for a University Library System. Assume that the following requirements were collected:

- The library contains a collection of RESOURCES. Each RESOURCE has a unique ID, a Title, an Acquisition_Date, and a Status (available, checked out, reserved, or under maintenance). Resources are categorized in several ways, as discussed below.
- RESOURCES are categorized based on their type. There are four types—BOOKS, PERIODICALS, DIGITAL_MEDIA, and EQUIPMENT.
- A BOOK has ISBN, Author, Publisher, and Publication_Year attributes. A PERIODICAL has ISSN, Frequency, and Publisher attributes. DIGITAL_MEDIA has Format and Duration attributes. EQUIPMENT has Model, Manufacturer, and Purchase_Date attributes.
- A RESOURCE can also be either CIRCULATING (can be checked out) or REFERENCE_ONLY (used only within the library).
- PUBLISHERs produce RESOURCES. The library keeps track of PUBLISHER information: Name, Address, Contact_Person, and Phone_Number. The Name is assumed to be unique.
- The library organizes EVENTS, each having a Name, Date, Time, and Location. EVENTS are related to all the RESOURCES that are featured or used during the event.
- To borrow resources, a PATRON needs to be registered. TRANSACTION confirms borrowing or returning of a RESOURCE by a PATRON.
- PATRONs can recommend resources to other PATRONs. In that case, the recommendation message and date are recorded.

Draw an EER diagram for this application. There must be a weak entity, at least one relationship attribute, at least one m:n relationship and at least one recursive relationship. Discuss any assumptions you make, and then justify your EER design choices (if required)